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Semiconductor structure and manufacturing method thereof

Abstract

The present disclosure relates to the technical field of semiconductors, and proposes a semiconductor structure and a manufacturing method thereof. The semiconductor structure includes a base and a communication portion, the communication portion being located in the base and including a first connection layer, a second connection layer, and a third connection layer, the second connection layer being located on the first connection layer, and the third connection layer being located on the second connection layer; wherein the first connection layer, the second connection layer, and the third connection layer include different conductive materials, and thermal expansion coefficients of the second connection layer and the third connection layer are both less than a thermal expansion coefficient of the first connection layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation of International Application No. PCT/CN2021/112594, filed on Aug. 13, 2021, which claims the priority to Chinese Patent Application 202110265111.5, titled “SEMICONDUCTOR STRUCTURE AND MANUFACTURING METHOD THEREOF” and filed on Mar. 10, 2021. The entire contents of International Application No. PCT/CN2021/112594 and Chinese Patent Application 202110265111.5 are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to the technical field of semiconductors, and in particular to a

semiconductor structure and a manufacturing method thereof.

BACKGROUND

(2) In the prior art, a metal conductive material filled in a communication portion may affect the lattice around the communication portion during thermal expansion, thus affecting the performance of the semiconductor structure.

SUMMARY

(3) The present disclosure provides a semiconductor structure and a manufacturing method thereof.

(4) According to a first aspect of the present disclosure, a semiconductor structure is provided, including: a base; a communication portion, the communication portion being located in the base and the communication portion including a first connection layer, a second connection layer, and a third connection layer, the second connection layer being located on the first connection layer, and the third connection layer being located on the second connection layer; wherein the first connection layer, the second connection layer, and the third connection layer include different conductive materials, and thermal expansion coefficients of the second connection layer and the third connection layer are both less than a thermal expansion coefficient of the first connection layer.

(5) According to a second aspect of the present disclosure, a semiconductor structure is provided, including: a base; a communication portion, the communication portion being located in the base and the communication portion including a first connection layer, a second connection layer, and a third connection layer, the second connection layer being located on the first connection layer, and the third connection layer being located on the second connection layer; wherein the second connection layer and the third connection layer both include graphene, and thermal expansion coefficients of the second connection layer and the third connection layer are both less than a thermal expansion coefficient of the first connection layer.

(6) According to a third aspect of the present disclosure, a method of manufacturing a semiconductor structure is provided, including: providing a matrix; and forming a communication portion on the matrix, the communication portion including a first connection layer, a second connection layer, and a third connection layer, the second connection layer being formed on the first connection layer, and the third connection layer being formed on the second connection layer; wherein the first connection layer, the second connection layer, and the third connection layer include different conductive materials, and thermal expansion coefficients of the second connection layer and the third connection layer are both less than a thermal expansion coefficient of the first connection layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The preferred embodiments of the present disclosure are described in detail below with reference to the accompanying drawings to make the objectives, features and advantages of the present disclosure more obvious. The drawings are merely exemplary illustrations of the present disclosure, and are not necessarily drawn to scale. The same reference numerals in the drawings always represent the same parts. In the drawings:

(2) FIG. 1 is a schematic structural diagram of a semiconductor structure according to a first exemplary implementation;

(3) FIG. 2 is a schematic structural diagram of a communication portion of the semiconductor structure according to the first exemplary implementation;

(4) FIG. 3 is a schematic structural diagram of a semiconductor structure according to a second exemplary implementation;

(5) FIG. 4 is a schematic structural diagram of a communication portion of the semiconductor

structure according to the second exemplary implementation;

(6) FIG. 5 is a schematic structural diagram of a semiconductor structure according to a third exemplary implementation;

(7) FIG. 6 is a schematic flowchart of a method of manufacturing a semiconductor structure according to an exemplary implementation;

(8) FIG. 7 is a schematic structural diagram of forming a hole in a method of manufacturing a semiconductor structure according to a first exemplary implementation;

(9) FIG. 8 is a schematic structural diagram of forming a hole in a second insulation layer in the method of manufacturing a semiconductor structure according to the first exemplary implementation;

(10) FIG. 9 is a schematic structural diagram of forming a hole in a third insulation layer in the method of manufacturing a semiconductor structure according to the first exemplary implementation;

(11) FIG. 10 is a schematic structural diagram of forming a first initial connection layer in the method of manufacturing a semiconductor structure according to the first exemplary implementation;

(12) FIG. 11 is a schematic structural diagram of forming a second initial connection layer in the method of manufacturing a semiconductor structure according to the first exemplary implementation;

(13) FIG. 12 is a schematic structural diagram of forming a second connection layer in the method of manufacturing a semiconductor structure according to the first exemplary implementation;

(14) FIG. 13 is a schematic structural diagram of forming a third initial connection layer in the method of manufacturing method of a semiconductor structure according to the first exemplary implementation;

(15) FIG. 14 is a schematic structural diagram of forming a third connection layer in the method of manufacturing a semiconductor structure according to the first exemplary implementation;

(16) FIG. 15 is a schematic structural diagram of forming a first initial connection layer in a method of manufacturing a semiconductor structure according to a second exemplary implementation;

(17) FIG. 16 is a schematic structural diagram of forming a second initial connection layer in the method of manufacturing a semiconductor structure according to the second exemplary implementation;

(18) FIG. 17 is a schematic structural diagram of forming a second connection layer in the method of manufacturing a semiconductor structure according to the second exemplary implementation;

(19) FIG. 18 is a schematic structural diagram of forming a third initial connection layer in the method of manufacturing a semiconductor structure according to the second exemplary implementation; and

(20) FIG. 19 is a schematic structural diagram of forming a third connection layer in the method of manufacturing a semiconductor structure according to the second exemplary implementation.

DETAILED DESCRIPTION

(21) The typical embodiments embodying the features and advantages of the present disclosure are described in detail below. It should be understood that the present disclosure may have various changes in different embodiments, which do not depart from the scope of the present disclosure. The description and drawings herein are essentially used for the purpose of explanation, rather than limiting the present disclosure.

(22) Different exemplary embodiments of the present disclosure are described below with reference to the accompanying drawings. The accompanying drawings form a part of the present disclosure, which show by way of example different exemplary structures, systems and steps that can implement various aspects of the present disclosure. It should be understood that other specific solutions of components, structures, exemplary devices, systems and steps may be used, and

structural and functional modifications may be made without departing from the scope of the present disclosure. Moreover, although the terms such as “above”, “between” and “within” may be used in this specification to describe different exemplary features and elements of the present disclosure, these terms are used herein only for convenience of description, for example, according to the directions of the examples in the drawings. Nothing in this specification should be understood as requiring a specific three-dimensional direction of the structure to fall within the scope of the present disclosure.

(23) An embodiment of the present disclosure provides a semiconductor structure. Referring to FIG. 1 to FIG. 5, the semiconductor structure includes: a base **10**; and a communication portion **21**, the communication portion **21** being located in the base **10**, the communication portion **21** including a first connection layer **211**, a second connection layer **212**, and a third connection layer **213**, the second connection layer **212** being located on the first connection layer **211**, and the third connection layer **213** being located on the second connection layer **212**; wherein the first connection layer **211**, the second connection layer **212**, and the third connection layer **213** include different conductive materials; thermal expansion coefficients of the second connection layer **212** and the third connection layer **213** are both less than a thermal expansion coefficient of the first connection layer **211**.

(24) The semiconductor structure in an embodiment of the present disclosure includes a base **10** and a communication portion **21**, wherein the communication portion **21** includes a first connection layer **211**, a second connection layer **212**, and a third connection layer **213** that are made of different conductive materials; thermal expansion coefficients of the second connection layer **212** and the third connection layer **213** are both less than a thermal expansion coefficient of the first connection layer **211**. That is, the second connection layer **212** and the third connection layer **213** change little under thermal expansion, thereby reducing the impact on the lattice around the communication portion **21**, thus improving the performance of the semiconductor structure.

(25) It should be noted that, since the thermal expansion coefficients of the second connection layer **212** and the third connection layer **213** are both less than the thermal expansion coefficient of the first connection layer **211**, a material with a low thermal expansion coefficient is used as part of the conductive materials of the communication portion **21**, to reduce the impact of metal thermal expansion on an active element region, and also reduce electrical changes of MOS elements in the peripheral lattice, thus reducing the size of the chip design. The second connection layer **212** and the third connection layer **213**, which have lower thermal expansion coefficients, are deformed less under heat, thus avoiding squeezing the lattice around the communication portion **21** with a high pressure, thereby avoiding the impact on other components in the base **10**.

(26) In some embodiments, the communication portion **21** is a part of a conductive portion **20**. The conductive portion **20** further includes a first conductive layer **22**. The first conductive layer **22** is located above the communication portion **21**, and the third connection layer **213** is connected to the first conductive layer **22**. Because the second connection layer **212** and the third connection layer **213** are close to the first conductive layer **22**, and the thermal expansion coefficients of the second connection layer **212** and the third connection layer **213** are relatively low, the first conductive layer **22** is not excessively squeezed.

(27) In some embodiments, the communication portion **21** is a through-silicon via.

(28) In an embodiment, the thermal expansion coefficient of the second connection layer **212** is less than the thermal expansion coefficient of the third connection layer **213**. That is, in the communication portion **21**, the conductive material with the lowest thermal expansion coefficient is located in the middle position. The base **10** is generally provided with a large number of components at a position corresponding to this middle position. Therefore, the conductive material with the lowest thermal expansion coefficient is subject to the smallest impact of thermal expansion, and basically will not squeeze components.

(29) In an embodiment, the second connection layer **212** or the third connection layer **213** includes

graphene, and the first connection layer **211** includes copper. Copper has a higher thermal expansion coefficient. However, since copper is arranged at the bottom, it does not affect the components in the base **10**. The second connection layer **212** or the third connection layer **213** is closer to the components in the base **10**, and graphene has a negative thermal expansion coefficient. Therefore, graphene can offset the thermal expansion of copper, thus avoiding the problem of squeezing.

(30) In some embodiments, the first connection layer **211** is made of copper, the second connection layer **212** is made of graphene, and the third connection layer **213** is made of tungsten.

(31) It should be noted that, when the second connection layer **212** includes graphene, which is sandwiched between the first connection layer **211** and the third connection layer **213**, graphene that has a certain thermal storage function can absorb heat from the conductive materials with the higher thermal expansion coefficients, that is, absorb heat from the first connection layer **211** and the third connection layer **213**, thereby reducing thermal expansion of other conductive materials.

(32) When the second connection layer **212** includes graphene, heat generated by the first connection layer **211** and the second connection layer **212** can be transferred to the first conductive layer **22** via graphene in time, and transmitted out through a structure connected to the first conductive layer **22**, thereby achieving the effect of rapid heat dissipation.

(33) In an embodiment, as shown in FIG. 1 and FIG. 2, the second connection layer **212** is located in the first connection layer **211**, that is, the second connection layer **212** is peripherally surrounded by the first connection layer **211**. The first connection layer **211** with the higher thermal expansion coefficient can protect the second connection layer **212**. However, due to the lower thermal expansion coefficient of the second connection layer **212**, the second connection layer **212** will not generate a large squeezing force due to thermal expansion.

(34) In some embodiments, the second connection layer **212** is located in the first connection layer **211**; the bottom of the third connection layer **213** is flush with the top of the second connection layer **212** and the top of the first connection layer **211**.

(35) In some embodiments, the second connection layer **212** is located in the first connection layer **211**, and the third connection layer **213** is located in the first connection layer **211**. That is, the second connection layer **212** and the third connection layer **213** are both peripherally surrounded by the first connection layer **211**, specifically as shown in FIG. 1 and FIG. 2.

(36) In an embodiment, as shown in FIG. 1 and FIG. 2, in a direction perpendicular to the base **10**, projection of the second connection layer **212** coincides with projection of the third connection layer **213**, that is, a cross sectional area of the second connection layer **212** is equal to a cross section of the third connection layer **213**.

(37) Optionally, the top of the third connection layer **213** is flush with the top of the first connection layer **211**. In this case, a groove is formed in the first connection layer **211**. The second connection layer **212** and the third connection layer **213** are located in the groove, and the top of the first connection layer **211** and the top of the third connection layer **213** are connected to the first conductive layer **22**, specifically as shown in FIG. 1 and FIG. 2. In this embodiment, the first connection layer **211**, the second connection layer **212**, and the third connection layer **213** include different conductive materials. The thermal expansion coefficient of the second connection layer **212** is less than the thermal expansion coefficient of the third connection layer **213**, and the thermal expansion coefficient of the third connection layer **213** is less than the thermal expansion coefficient of the first connection layer **211**. The first connection layer **211** is made of copper, the second connection layer **212** is made of graphene, and the third connection layer **213** is made of tungsten.

(38) In an embodiment, as shown in FIG. 3 and FIG. 4, in the direction perpendicular to the base **10**, projection of the first connection layer **211**, projection of the second connection layer **212**, and projection of the third connection layer **213** coincide with each other. That is, the first connection layer **211**, the second connection layer **212**, and the third connection layer **213** are sequentially

stacked along a height direction of the base **10**, and the cross sectional area of the first connection layer **211**, the cross sectional area of the second connection layer **212**, and the cross sectional area of the third connection layer **213** are equal. In this case, only the third connection layer **213** is directly connected to the first conductive layer **22**. In this embodiment, the first connection layer **211**, the second connection layer **212**, and the third connection layer **213** include different conductive materials. The thermal expansion coefficient of the second connection layer **212** is less than the thermal expansion coefficient of the third connection layer **213**, and the thermal expansion coefficient of the third connection layer **213** is less than the thermal expansion coefficient of the first connection layer **211**. The first connection layer **211** is made of copper, the second connection layer **212** is made of graphene, and the third connection layer **213** is made of tungsten.

(39) In some embodiments, as shown in FIG. 5, the second connection layer **212** is located in the first connection layer **211**, and in the direction perpendicular to the base **10**, projection of the first connection layer **211** coincides with projection of the third connection layer **213**. That is, the bottom of the third connection layer **213** is flush with the top of the second connection layer **212** and the top of the first connection layer **211**.

(40) In an embodiment, as shown in FIG. 1, FIG. 3 and FIG. 5, the base **10** includes: a silicon substrate **11**, the bottom of the second connection layer **212** being lower than an upper surface of the silicon substrate **11**; and an insulation layer **12**, the insulation layer **12** covers the upper surface of the silicon substrate **11** and covers an outer surface of the communication portion **21**, and the first conductive layer **22** is located in the insulation layer **12**.

(41) Specifically, the base **10** includes the silicon substrate **11** and the insulation layer **12**. The communication portion **21** is partially located in the silicon substrate **11**; the bottom of the second connection layer **212** is lower than the upper surface of the silicon substrate **11**, and the thermal expansion coefficient of the second connection layer **212** is less than the thermal expansion coefficient of the first connection layer **211**. Therefore, the second connection layer **212** will not be expanded excessively under heat, thereby avoiding the impact on a part of the silicon substrate **11** which is close to the upper surface of the second connection layer **212**.

(42) In an embodiment, a functional layer **111** is formed in the silicon substrate **11**, and the bottom of the second connection layer **212** is lower than a lower surface of the functional layer **111**. That is, a total height of the second connection layer **212** and the third connection layer **213** is not less than a height of the functional layer **111** in the silicon substrate **11**, thereby avoiding squeezing the functional layer **111** under thermal expansion. The height of the second connection layer **212** may be greater than the height of the functional layer **111**.

(43) It should be noted that various components may be provided in the functional layer **111**. Types of the components are not limited herein, and can be selected according to the actual needs. The focus herein is that the second connection layer **212**, after overall thermal expansion, will not squeeze the functional layer **111**, thereby protecting the functional layer **111**.

(44) In an embodiment, the top of the second connection layer **212** is higher than the upper surface of the silicon substrate **11**. That is, the height of the second connection layer **212** may be greater than the height of the functional layer **111**, thereby protecting the part of the silicon substrate **11** which is close to the insulation layer **12**, and avoiding squeezing the functional layer **111**.

(45) In some embodiments, the first connection layer **211**, the second connection layer **212**, and the third connection layer **213** may be columnar structures.

(46) In some embodiments, the second connection layer **212** may have a diameter of 200 nm to 10 μm and a depth of 1 μm to 20 μm . A portion of the second connection layer **212** which enters the silicon substrate **11** has a depth greater than or equal to 1 μm .

(47) Specifically, the silicon substrate **11** may be made of a silicon-containing material. The silicon substrate **11** may be made of any suitable material, including, for example, at least one of silicon, monocrystalline silicon, amorphous silicon, silicon-germanium, monocrystalline silicon-germanium, silicon carbide, or sapphire.

(48) The insulation layer **12** may include silicon dioxide (SiO₂), silicon oxide carbon (SiOC), silicon nitride (SiN), silicon carbon nitride (SiCN), and other related integrated-circuit insulation materials.

(49) In an embodiment, as shown in FIG. 1, FIG. 3 and FIG. 5, the semiconductor structure further includes: a second conductive layer **30**, the second conductive layer **30** being located in the base **10**, the second conductive layer **30** is spaced apart from the first conductive layer **22** and is located above the first conductive layer **22**, the first conductive layer **22** is connected to the second conductive layer **30** through connection pillars **31**, and the connection pillars **31** are conductive structures.

(50) An embodiment of the present disclosure further provides a semiconductor structure, including: a base **10**; and a communication portion **21**, the communication portion **21** being located in the base **10**, the communication portion **21** includes a first connection layer **211**, a second connection layer **212**, and a third connection layer **213**, the second connection layer **212** being located on the first connection layer **211**, and the third connection layer **213** being located on the second connection layer **212**; the second connection layer **212** and the third connection layer **213** both include graphene, and thermal expansion coefficients of the second connection layer **212** and the third connection layer **213** are both less than a thermal expansion coefficient of the first connection layer **211**.

(51) Compared with the foregoing embodiment, the semiconductor structure in this embodiment includes the first connection layer **211** and graphene located above the first connection layer **211**, graphene can effectively avoid impact on the lattice around the communication portion **21**, and heat can be transmitted out in time through graphene. In this way, the performance of the semiconductor structure is improved.

(52) It should be noted that, reference may be made to the foregoing embodiment for other related structures, and details are not described herein.

(53) An embodiment of the present disclosure further provides a method of manufacturing a semiconductor structure. Referring to FIG. 6, the method of manufacturing a semiconductor structure includes: **S101**: Provide a matrix **13**. **S103**: Form a communication portion **21** on the matrix **13**, the communication portion **21** including a first connection layer **211**, a second connection layer **212**, and a third connection layer **213**, the second connection layer **212** being formed on the first connection layer **211**, and the third connection layer **213** being formed on the second connection layer **212**.

(54) The first connection layer **211**, the second connection layer **212**, and the third connection layer **213** include different conductive materials; thermal expansion coefficients of the second connection layer **212** and the third connection layer **213** are both less than a thermal expansion coefficient of the first connection layer **211**.

(55) According to the method of manufacturing a semiconductor structure in an embodiment of the present disclosure, a communication portion **21** is formed on a matrix **13**, the communication portion **21** including a first connection layer **211**, a second connection layer **212**, and a third connection layer **213** that are made of different conductive materials; thermal expansion coefficients of the second connection layer **212** and the third connection layer **213** are both less than a thermal expansion coefficient of the first connection layer **211**, and the third connection layer **213** is connected to a first conductive layer **22**. That is, the second connection layer **212** and the third connection layer **213** change little under thermal expansion, thereby reducing the impact on the lattice around the communication portion **21**, thus improving the performance of the semiconductor structure.

(56) In an embodiment, the second connection layer **212** or the third connection layer **213** includes graphene, and the first connection layer **211** includes copper. Copper has a higher thermal expansion coefficient. However, since copper is arranged at the bottom, it does not affect the components of the base **10**. The second connection layer **212** or the third connection layer **213** is

closer to the components in the base **10**, and graphene has a negative thermal expansion coefficient. Therefore, graphene can offset the thermal expansion of copper, thus avoiding the problem of squeezing.

(57) In an embodiment, an opening **14** is formed in the first connection layer **211**; the second connection layer **212** and the third connection layer **213** are sequentially formed in the opening **14**, that is, the structure shown in FIG. **1** and FIG. **2** is formed.

(58) Specifically, as shown in FIG. **7**, the matrix **13** includes a silicon substrate **11** and a first insulation layer **15**. A hole **16** is formed in the silicon substrate **11** and the first insulation layer **15**; the hole **16** penetrates the first insulation layer **15** but does not necessarily penetrate the silicon substrate **11**, which is not limited herein.

(59) On the basis of FIG. **7**, a second insulation layer **17** is covered on the first insulation layer **15**, and the second insulation layer **17** covers the wall of the hole **16**, as shown in FIG. **8**.

(60) On the basis of FIG. **8**, a third insulation layer **18** is covered on the second insulation layer **17**, and the third insulation layer **18** covers an upper surface and side surfaces of the second insulation layer **17**, as shown in FIG. **9**.

(61) On the basis of FIG. **9**, a first initial connection layer **19** is covered on the third insulation layer **18**; the first initial connection layer **19** fills part of the hole **16**, and an opening **14** is formed above the first initial connection layer **19**, as shown in FIG. **10**

(62) On the basis of FIG. **10**, a second initial connection layer **32** is covered on the first initial connection layer **19**, and the second initial connection layer **32** fills the opening **14**, as shown in FIG. **11**.

(63) On the basis of FIG. **11**, a part of the second initial connection layer **32** which covers an upper surface of the first initial connection layer **19** and is located on an outer side of the opening **14** is removed, and the opening **14** is partially exposed, to form the second connection layer **212**, as shown in FIG. **12**.

(64) On the basis of FIG. **12**, a third initial connection layer **33** is covered on the first initial connection layer **19**, and the third initial connection layer **33** fills the opening **14**, as shown in FIG. **13**.

(65) On the basis of FIG. **13**, the third initial connection layer **33** which covers the upper surface of the first initial connection layer **19** and is located on the outer side of the opening **14**, and a part of the first initial connection layer **19** which is located on the third insulation layer **18** are removed, to form the first connection layer **211** and the third connection layer **213**, as shown in FIG. **14**. That is, the semiconductor structure shown in FIG. **1** can be finally formed.

(66) In an embodiment, the first connection layer **211**, the second connection layer **212**, and the third connection layer **213** are sequentially formed in the matrix **13**. In a direction perpendicular to the matrix **13**, projection of the first connection layer **211**, projection of the second connection layer **212**, and projection of the third connection layer **213** coincide with each other. That is, the structure shown in FIG. **3** and FIG. **4** is formed.

(67) Specifically, on the basis of FIG. **10**, part of the first initial connection layer **19** is removed, that is, an area above the hole **16** is exposed, to form the first connection layer **211**, as shown in FIG. **15**.

(68) On the basis of FIG. **15**, a second initial connection layer **32** is covered on the third insulation layer **18**, and the second initial connection layer **32** fills the hole **16**, as shown in FIG. **16**.

(69) On the basis of FIG. **16**, the second initial connection layer **32** is partially removed, that is, an area above the hole **16** is exposed, to form the second connection layer **212**, as shown in FIG. **17**.

(70) On the basis of FIG. **17**, a third initial connection layer **33** is covered on the third insulation layer **18**, and the third initial connection layer **33** fills the hole **16**, as shown in FIG. **18**.

(71) On the basis of FIG. **18**, the third initial connection layer **33** is partially removed, that is, an upper surface of the third initial connection layer **33** is made to be flush with an upper surface of the third insulation layer **18**, to form the third connection layer **213**, as shown in FIG. **19**. That is,

the semiconductor structure shown in FIG. 3 can be finally formed.

(72) It should be noted that, the first insulation layer **15**, the second insulation layer **17**, and the third insulation layer **18** may include silicon dioxide (SiO_2), silicon oxide carbon (SiOC), silicon nitride (SiN), silicon carbon nitride (SiCN), and other related integrated-circuit insulation materials.

(73) It should be noted that, a formation process of the first insulation layer **15**, the second insulation layer **17**, the third insulation layer **18**, the second initial connection layer **32**, and the third initial connection layer **33** can be selected from a group consisting of a physical vapor deposition (PVD) process, a chemical vapor deposition (CVD) process, an atomic layer deposition (ALD) process, an in-situ steam generation (ISSG) process, and a spin on dielectric (SOD) process, etc., which is not limited herein.

(74) The formation process of the opening **14** and the hole **16** includes lithography and etching. After each coating is formed, a chemical mechanical polishing (CMP) process can be performed to ensure the flatness of the coating. The first initial connection layer **19** can be formed by a process such as electroplating or sputtering, which is not limited therein.

(75) In an embodiment, the semiconductor structure can be formed by using the foregoing method of manufacturing a semiconductor structure. The method of manufacturing a semiconductor structure can further include forming a second conductive layer **30**, and the like, which is not limited herein. The silicon substrate **11** and the various insulation layers described above are all part of the base **10**, that is, the matrix **13** is a part of the base **10**. The first insulation layer **15**, the second insulation layer **17**, and the third insulation layer **18** are all part of the insulation layer **12**.

(76) A person skilled in the art may easily think of other implementation solutions of the present disclosure after considering the specification and practicing the content disclosed herein. The present disclosure is intended to cover any variations, purposes or applicable changes of the present disclosure. Such variations, purposes or applicable changes follow the general principle of the present disclosure and include common knowledge or conventional technical means in the technical field which is not disclosed in the present disclosure. The specification and implementations are merely considered as illustrative, and the real scope and spirit of the present disclosure are directed by the appended claims.

(77) It should be noted that, the present disclosure is not limited to the precise structures that have been described above and shown in the accompanying drawings, and can be modified and changed in many ways without departing from the scope of the present disclosure. The scope of the present disclosure is defined by the appended claims.

Claims

1. A semiconductor structure, comprising: a base comprising a silicon substrate; a communication portion, the communication portion being located in the base and the communication portion comprising a first connection layer, a second connection layer, and a third connection layer, the second connection layer being located on the first connection layer, and the third connection layer being located on the second connection layer; wherein a top of the second connection layer is higher than an upper surface of the silicon substrate, a bottom of the second connection layer is lower than an upper surface of the silicon substrate; wherein a sum of a heights of the second connection layer and the third connection layer is less than one half of a height of the first connection layer; and wherein a thermal expansion coefficient of the second connection layer is less than a thermal expansion coefficient of the third connection layer, and the thermal expansion coefficient of the third connection layer is less than a thermal expansion coefficient of the first connection layer; the first connection layer is made of copper, the second connection layer is made of graphene, and the third connection layer is made of tungsten.

2. The semiconductor structure according to claim 1, wherein the second connection layer is

located in the first connection layer.

3. The semiconductor structure according to claim 2, wherein the third connection layer is located in the first connection layer.

4. The semiconductor structure according to claim 3, wherein in a direction perpendicular to the base, a projection of the second connection layer coincides with a projection of the third connection layer; wherein a top of the third connection layer is flush with a top of the first connection layer.

5. The semiconductor structure according to claim 1, further comprising: an insulation layer, the insulation layer covering the upper surface of the silicon substrate and covering an outer surface of the communication portion.

6. The semiconductor structure according to claim 5, wherein a functional layer is formed in the silicon substrate, and the bottom of the second connection layer is lower than a lower surface of the functional layer.

7. The semiconductor structure according to claim 5, wherein a portion of the second connection layer entering the silicon substrate has a depth greater than or equal to 1 μm .

8. The semiconductor structure according to claim 1, wherein the second connection layer has a diameter of 200 nm to 10 μm , and the second connection layer has a depth of 1 μm to 20 μm .

9. A method for forming the semiconductor structure of claim 1, comprising: providing a base comprising a silicon substrate; forming a communication portion on the base, the communication portion comprising a first connection layer, a second connection layer, and a third connection layer, the second connection layer being formed on the first connection layer, and the third connection layer being formed on the second connection layer; wherein a top of the second connection layer is higher than an upper surface of the silicon substrate, a bottom of the second connection layer is lower than an upper surface of the silicon substrate; wherein a sum of a heights of the second connection layer and the third connection layer is less than one half of a height of the first connection layer; and wherein a thermal expansion coefficient of the second connection layer is less than a thermal expansion coefficient of the third connection layer, and the thermal expansion coefficient of the third connection layer is less than a thermal expansion coefficient of the first connection layer; the first connection layer is made of copper, the second connection layer is made of graphene, and the third connection layer is made of tungsten.

10. The method according to claim 9, wherein an opening is formed in the first connection layer, and the second connection layer and the third connection layer are sequentially formed in the opening.

11. The method according to claim 9, wherein the first connection layer, the second connection layer, and the third connection layer are sequentially formed in the base; wherein in a direction perpendicular to the base, a projection of the second connection layer coincides with, a projection of the third connection layer.
